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# Research on Building a Web Full-Stack Development Talent Training System Based on Job Requirements

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# Research on Building a Web Full-Stack Development Talent Training System Based on Job Requirements

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## Abstract

The demand for Web full-stack development talents has surged, but the existing training system has problems such as lagging courses and inconsistent evaluation standards, which leads to a disconnect between talent capabilities and job requirements. Therefore, this article first analyzes the job requirements for Web full-stack development, including industry technology trends, corporate recruitment data and core competency dimensions; then diagnoses the problems of the existing training system, involving colleges and universities, training institutions, and corporate internal training; finally, it constructs a training program based on job requirements, covering basic competency matrix, advanced modules, practical training models and other contents.

## CCS Concepts

• **Information systems—Information systems applications—Decision support systems**;

## Keywords

Web full-stack development, talent training, job requirements, training system, industry-education integration

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## 1 Introduction

The demand for Web full-stack development positions continues to expand in digital transformation, the speed of industry technology iteration is accelerating, and companies have higher requirements for the breadth and depth of talent skills. In the existing training system, college course updates lag behind technological development, training institutions focus on short-term effects, corporate internal training resources are limited, learning paths lack systematicity, and competency assessment standards are inconsistent. These factors lead to a gap between talent supply and actual job requirements. It is necessary to start from job requirements, re-sort out the training logic, build a talent training system that adapts to industry development, and solve the problem of mismatch between talent capabilities and job requirements.

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## 2 Analysis of job requirements for Web full-stack development

### 2.1 Research on industry technology development trends

The popularity of cloud computing has prompted Web applications to gradually migrate to the cloud, and many companies have used cloud services to achieve flexible deployment and elastic resource expansion. Containerization technologies such as Docker and Kubernetes have further optimized the operating environment and management efficiency of applications. In the field of big data, Web full-stack development needs to connect to the big data platform to achieve data visualization and interaction. Artificial intelligence technology is constantly being integrated into Web applications, such as the implementation of intelligent customer service and personalized recommendations, requiring developers to master the relevant algorithm interface calls and data processing capabilities. In the era of the Internet of Things, the Web end is an important interface for the management and monitoring of IoT devices. Full-stack developers need to have the technical capabilities of device communication protocol parsing and data security transmission. 5G communication technology has promoted the development of Web applications in real-time interaction, high-definition video streaming transmission, etc., and has put forward higher requirements for the performance optimization of Web full-stack development [1]. At the same time, with the diversification of network security threats, Web full-stack developers must keep up with the development of security technologies to ensure the security of applications in data transmission, storage and other links.

### 2.2 Data mining of enterprise recruitment needs

According to the data from major recruitment platforms, the demand for Web full-stack development talents by enterprises continues to grow. In terms of education, the proportion of undergraduates and above is relatively high, reflecting the company's requirements for the integrity of the talent knowledge system. In terms of work experience, there is a prominent demand for talents with 3-5 years of experience. Such personnel have a certain amount of project practice accumulation and can quickly adapt to the rhythm of enterprise project development. In terms of skill requirements, proficiency in at least one mainstream front-end framework such as React, Vue or Angular, as well as back-end languages such as Node.js, Java, Python, etc. are basic requirements. At the same time, familiarity with database operations, including relational databases

**Table 1: Distribution of Core Skill Requirements for Web Full-Stack Development Positions. (Data Source: BOSS Zhipin (Q1-Q2 2025))**

| Dimension Category | Skill / Requirement Item | Frequency (%) |
|---|---|---|
| Education | Bachelor's degree or above | 82.6% |
| | College diploma | 15.2% |
| | Graduate degree | 2.2% |
| Work Experience | 3-5 years | 48.7% |
| | 1-3 years | 31.4% |
| | 5-10 years | 14.9% |
| Frontend Tech | React | 41.2% |
| | Vue | 37.8% |
| | Angular | 10.4% |
| Backend Language | Java | 45.7% |
| | Node.js | 32.5% |
| | Python | 28.1% |
| **Database** | MySQL | 69.5% |
| | PostgreSQL | 21.3% |
| | MongoDB | 25.4% |
| **Cloud Service** | Alibaba Cloud | 27.9% |
| | AWS | 18.6% |
| | Azure | 11.5% |
| DevOps Tools | Docker | 56.2% |
| | Kubernetes(K8s) | 38.4% |
| | CI/CD(GitLab/Jenkins) | 42.7% |

MySQL, PostgreSQL and non-relational databases MongoDB, understanding cloud services such as AWS, Azure, Alibaba Cloud, and mastering DevOps tools such as Docker, K8s, GitCI/CD, etc. have also become important considerations for corporate recruitment. In addition, talents with a good sense of code standards, familiarity with Git version management and agile development processes are more favored. Some companies also emphasize talents' ability to learn and explore new technologies and their team collaboration and communication skills, as shown in table 1.

### 2.3 Analysis of core competence dimensions

The core competence of Web full-stack development talents covers multiple dimensions. In terms of technical capabilities, the front-end needs to be proficient in HTML5 and CSS3 to build page structure and style, be proficient in using JavaScript for interactive logic writing, master the front-end framework to improve development efficiency and code maintainability, and understand the construction tools to achieve automated development processes. The back-end needs to master at least one development language to process server requests and responses, be familiar with the RESTful API design specifications to achieve front-end and back-end data interaction, and have the ability to design, manage and optimize databases. In terms of network knowledge, it is necessary to be familiar with the principles of the HTTP protocol, understand the network request and response process, and master the basic knowledge such as DNS and IP addresses. Version control capabilities are indispensable, and code version management is performed through Git to ensure the consistency and traceability of code during team collaborative development. In terms of soft skills, project management capabilities require developers to use methods such as agile development to reasonably plan project progress and coordinate resources; team collaboration capabilities enable developers to communicate and collaborate efficiently with product managers, designers and other developers to promote the smooth progress of projects; code review capabilities help improve code quality, follow industry best practices, and optimize the overall project architecture.

## 3 Diagnosis of Problems in the Existing Training System

### 3.1 Analysis of the Lagging Curriculum of Colleges and Universities

The updating speed of colleges and universities in the curriculum of Web full-stack development is far less than the iteration of the industry technology. The revision process of the syllabus is cumbersome, resulting in cutting-edge technologies such as emerging front-end frameworks, back-end high-performance development frameworks, etc., which are difficult to integrate in a timely manner. Traditional courses focus too much on theory, and do not pay enough attention to the practical aspects, such as database courses, more focused on theoretical explanations, students lack of database optimization, complex queries and other practical experience. In cloud computing, big data and other popular areas, the university curriculum system has not yet been constructed, students are difficult to systematically learn the relevant knowledge, resulting in

a disconnect between the knowledge structure and the current industry demand, graduates need to re-learn a lot of new knowledge, new skills after entering the enterprise.

### 3.2 The defects of commercialized courses of training institutions

In order to pursue commercial interests, the course contents of training institutions are mostly oriented to test-taking and quick results. Teaching content often emphasizes skills over principles, and although students can master some problem-solving routines or operational processes in the short term, they do not have a deep understanding of the underlying logic of the technology. The curriculum lacks systematic, the connection between the modules is not close, such as front-end and back-end courses, failed to effectively integrate into a coherent full-stack development knowledge system. Part of the organization to compress the cost, hired teachers lack of actual project experience, teaching process is difficult to integrate real project cases into the teaching, resulting in a disconnect between the knowledge learned by the students and the actual work scene, it is difficult to quickly adapt to the complexity of the changing project development needs after entering the post [2].

### 3.3 The limitations of enterprise training resources

Enterprise training is often subject to resource constraints, limited capital investment, difficult to hire top industry experts to teach, but also unable to purchase the latest technology and equipment to assist in teaching. Training schedule and staff daily work conflicts, employees are difficult to devote themselves to learning, resulting in poor training results. The content of internal training is mostly based on the current business needs, the lack of forward-looking guidance on the future development trend of the industry, the staff's knowledge horizons are limited. At the same time, internal training is often targeted at specific projects or technical points, the lack of Web full-stack development knowledge system of comprehensive grooming, employees are difficult to build a complete knowledge structure, which is not conducive to their long-term career development.

### 3.4 Lack of systematic learning path

Whether it is university education, training courses, or employee independent learning, there is a lack of systematic learning path. The overall learning plan lacks logic, and the learning objectives of each stage are not clear, for example, after learning the basics of Web development, students are not clear whether they should learn the front-end or the back-end, and how to combine the two organically. Lack of effective integration between different learning channels, college courses, training institutions and corporate training content in their own way, failing to form a unified and coherent learning chain, resulting in fragmentation of learner knowledge, it is difficult to integrate what they have learned, and unable to build a solid and systematic Web full-stack development knowledge and skills system.

### 3.5 Inconsistency of competency assessment standards

There is a lack of uniform and authoritative competency assessment standards in the cultivation of Web full-stack development talents. Colleges and universities assess students' ability by examination results and coursework, focusing on theoretical knowledge mastery and unable to comprehensively measure students' practical and comprehensive application ability. Training organizations mostly evaluate students by simulating the completion of projects, which is a single evaluation method, and there is a gap between project scenarios and real work scenarios. The evaluation of internal training is often linked to the performance of employees, which is easily interfered by subjective factors, and the evaluation results can hardly truly reflect the training effect and ability improvement of employees. This kind of non-uniform evaluation standard makes it difficult to accurately control the quality of talent training, and is not conducive to the flow and articulation of talents between different training systems.

## 4 Web full-stack developer talent cultivation construction program based on job requirements

### 4.1 Basic ability matrix construction

According to the cluster analysis, four dimensions of the job requirement specification model can be derived, corresponding to the four dimensions of Knowledge, Skill, Ability, and quality of qualifications in the Qualification for Occupation (KSAO) model. (Others) 4 dimensions. The keywords included in each dimension are shown in Figure 1. Determine the competency standard of each level by the formula of competency attainment value:

$$C_{ij} = \sum_{k=1}^{n} (S_{ijk} \times W_k) / \sum_{k=1}^{n} W_k$$

where $C_{ij}$ is the competency value of the jth tier of the ith dimension, $S_{ijk}$ is the score of the kth skill of the jth tier of the ith dimension, $W_k$ is the weight of the corresponding skill). Curriculum units are set up according to the matrix framework, and each unit corresponds to the skill points of a specific dimensional tier, and the $S_{ijk}$ value is obtained through the stage test, which is substituted into the formula to verify the achievement of competence, and the skill points that do not meet the standard are included in the supplemental training module [3], as shown in Figure 2.

Table 2 demonstrates the effect of different dimensions of skill scores and their weights on competency values.

### 4.2 Advanced Capability Module Design

Modules are divided by front-end framework integration, back-end architecture optimization, and full-stack project integration, and each module contains 3-4 Skill subsets. Adopt the Skill relevance calculation formula:

$$R_{mn} = \frac{\sum_{p=1}^{q} (A_{mp} \cap A_{np})}{\sum_{p=1}^{q} (A_{mp} \cup A_{np})}$$

where $R_{mn}$ is the relevance of the mth and the nth subset, $A_{mp}$ is the pth Skill of the mth subset, $A_{np}$ is the pth Skill of the nth

**Table 2: Statistical Analysis of Competency Qualification Threshold Values**

| Dimension | Level | Skill Item | Skill Score ($S_{ijk}$) | Skill Weight ($W_k$) | Competency Value Calculation ($C_{ij}$) | Level |
|---|---|---|---|---|---|---|
| **Knowledge** | Beginner | Skill A | 0.8 | 0.3 | 0.24 | **Knowledge** |
| | Intermediate | Skill B | 0.9 | 0.4 | 0.36 | |
| | Advanced | Skill C | 0.7 | 0.5 | 0.35 | |
| **Skills** | Beginner | Skill D | 0.6 | 0.2 | 0.12 | Skill |
| | Intermediate | Skill E | 0.8 | 0.3 | 0.24 | |
| | Advanced | Skill F | 0.9 | 0.4 | 0.36 | |
| **Abilities** | Beginner | Skill G | 0.7 | 0.1 | 0.07 | **Abilities** |
| | Intermediate | Skill H | 0.6 | 0.2 | 0.12 | |
| | Advanced | Skill I | 0.5 | 0.3 | 0.15 | |
| **Professionalism** | Beginner | Skill J | 0.9 | 0.1 | 0.09 | **Professionalism** |
| | Intermediate | Skill K | 0.8 | 0.2 | 0.16 | |
| | Advanced | Skill L | 0.7 | 0.3 | 0.21 | |

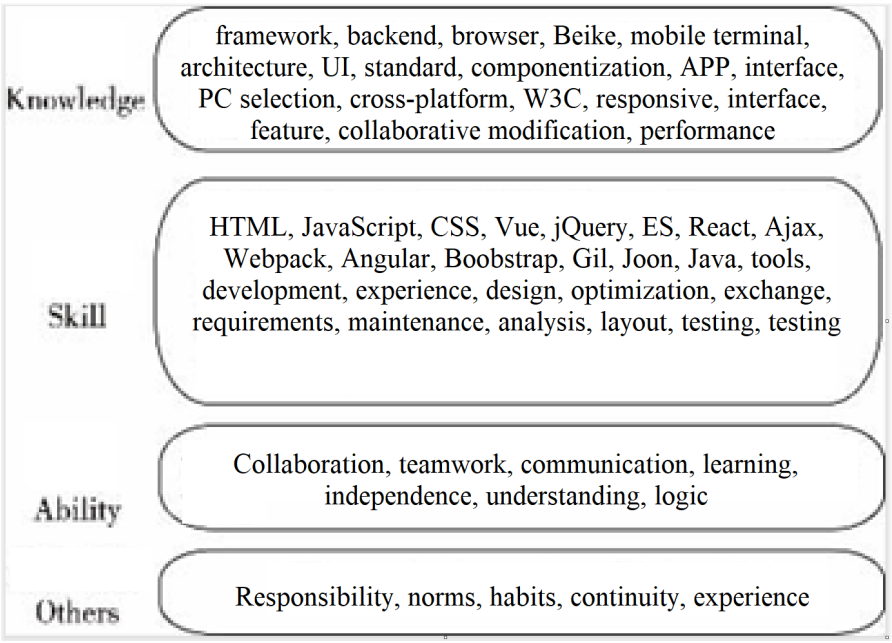


**Figure 1: Cluster analysis results**

subset) Determine the teaching of subset Order. Based on the R_mn value, the teaching schedule of Skill subsets is arranged, and the subsets with high relevance are taught consecutively, while the cross-module articulation courses are set to ensure the logical coherence among Skill subsets.

Table 3 shows the correlation between Skill subsets to help determine the teaching sequence and logical coherence between modules.

## 4.3 Project-driven practical training model

Establish the project complexity assessment model: D = $\alpha\times$ L + $\beta\times$ I + $\gamma\times$ T (where D is the project complexity, L is the number of technology stacks, I is the number of module interactions, T is the development cycle, $\alpha, \beta, \gamma$ is the weighting factor and $\alpha + \beta + \gamma = 1$). The practical training projects are divided into 3 levels according to the value of D. The primary project D<0.4, intermediate 0.4≤D≤0.7, and advanced D>0.7. Learners enter the next level after completing

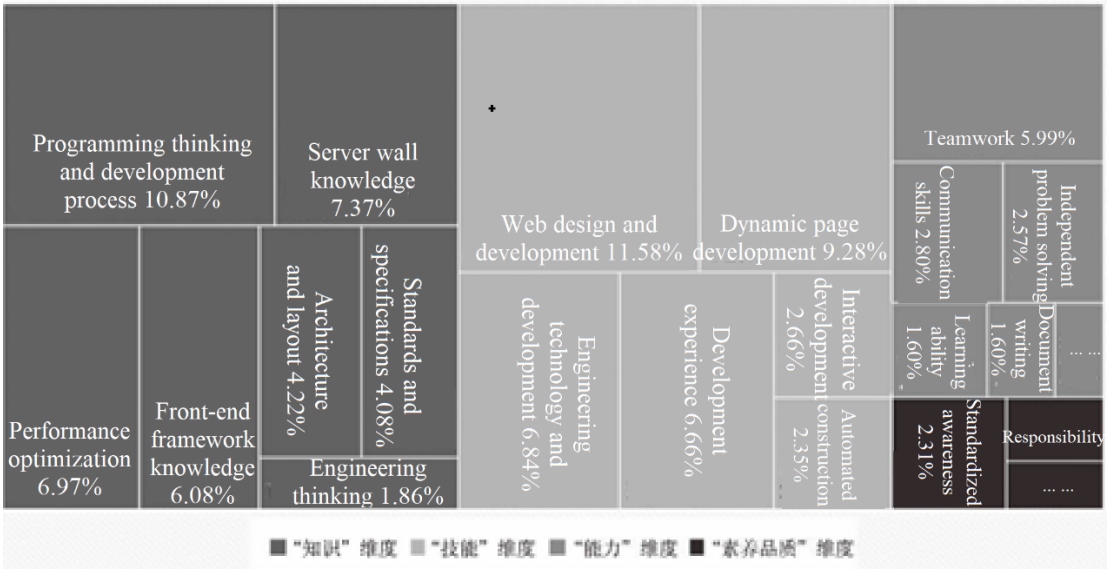


**Figure 2: Web front-end developer job demand characteristics model**

**Table 3: Statistical analysis of Skill's relevance**

| Subset ID | Skill Items | Intersection Count | Union Count | Relevance ($R_{mn}$) |
|---|---|---|---|---|
| m1 | A1, A2 | 2 | 4 | 0.5 |
| m2 | B1, B2 | 1 | 5 | 0.2 |
| m3 | C1, C2 | 3 | 3 | 1.0 |
| n1 | D1, D2 | 2 | 4 | 0.5 |
| n2 | E1, E2 | 1 | 5 | 0.2 |
| n3 | F1, F2 | 3 | 3 | 1.0 |

**Table 4: Statistical Analysis of Project Complexity Assessment**

| Project ID | Tech Stack Count (L) | Module Interactions (I) | Dev Cycle (T) | Weight Coefficients ($\alpha, \beta, \gamma$) | Project Complexity (D) | Project Level |
|---|---|---|---|---|---|---|
| Project 1 | 5 | 10 | 4 | $\alpha = 0.4, \beta = 0.3, \gamma = 0.3$ | 0.5 | Intermediate |
| Project 2 | 3 | 8 | 6 | $\alpha = 0.3, \beta = 0.4, \gamma = 0.3$ | 0.6 | Intermediate |
| Project 3 | 4 | 12 | 3 | $\alpha = 0.2, \beta = 0.5, \gamma = 0.3$ | 0.7 | Advanced |
| Project 4 | 2 | 5 | 8 | $\alpha = 0.5, \beta = 0.2, \gamma = 0.3$ | 0.3 | Entry-level |
| Project 5 | 6 | 15 | 2 | $\alpha = 0.1, \beta = 0.4, \gamma = 0.5$ | 0.8 | Advanced |

the project of the previous level and passing the assessment, and each project is set up with a clear documentation of the functional module division, which is required to be realized in accordance with the modularization development process, the daily submission of the code updating logs, and the weekly module integration test [5], as shown in Figure 3.

Table 4 shows the assessment results of different project complexity levels to help determine the practical training level of the project and the corresponding teaching strategies.

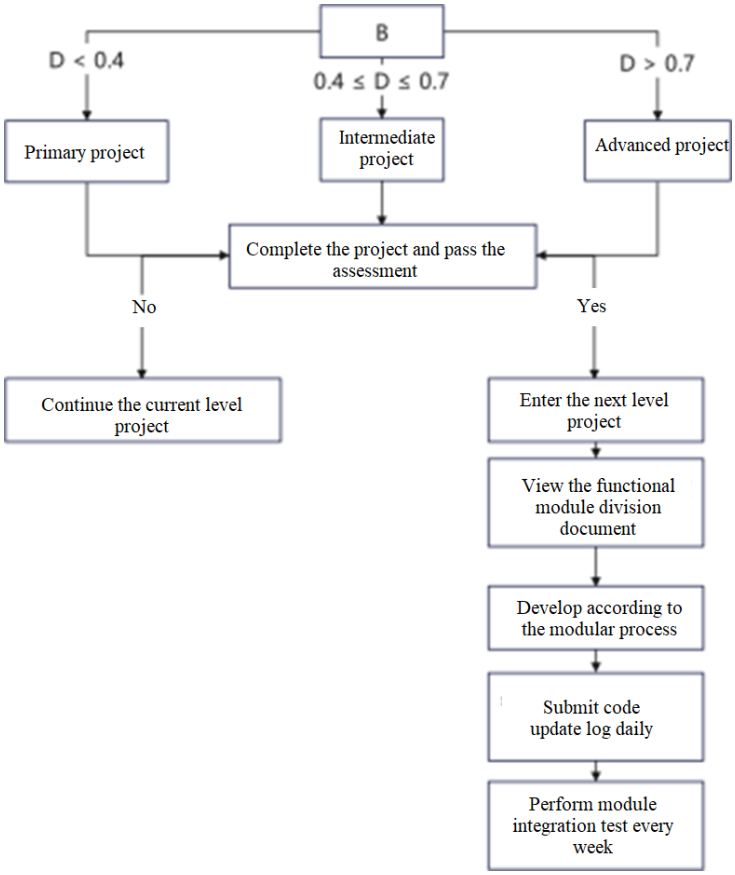


**Figure 3: Project-driven practical training flowchart**

**Table 5: Course update frequency statistics**

| Time Period (t) | Technology Iteration Volume ($\Delta V_t$) | Update Frequency Coefficient (F) |
|---|---|---|
| 1 | 5 | 0.05 |
| 2 | 10 | 0.1 |
| 3 | 15 | 0.15 |
| 4 | 20 | 0.2 |
| 5 | 25 | 0.25 |
| 6 | 30 | 0.3 |
| … | … | … |

## 4.4 Dynamic course update mechanism

Construct the formula for calculating the frequency of course update:

$$F = \frac{\sum_{t=1}^{m} \Delta V_t}{\sum_{t=1}^{m} t}$$

where F is the updating frequency coefficient, $\Delta V_t$ is the amount of technology iteration in the tth cycle, and t is the time period). Set the F threshold and initiate the course revision process when the actual calculated value exceeds the threshold [4]. Establish a technology radar database, collect real-time data from open source community version updates and enterprise technology stack changes, calculate the $\Delta V_t$ value every month and substitute it into the formula to generate a list of course update priorities. Revise the course units in the order of the list, delete the outdated technical content, supplement the new technical knowledge points, and synchronously update the supporting exercises and Project cases, as shown in Figure 4.

Table 5 demonstrates the impact of the amount of technology iterations in different cycles on the frequency of course updates.

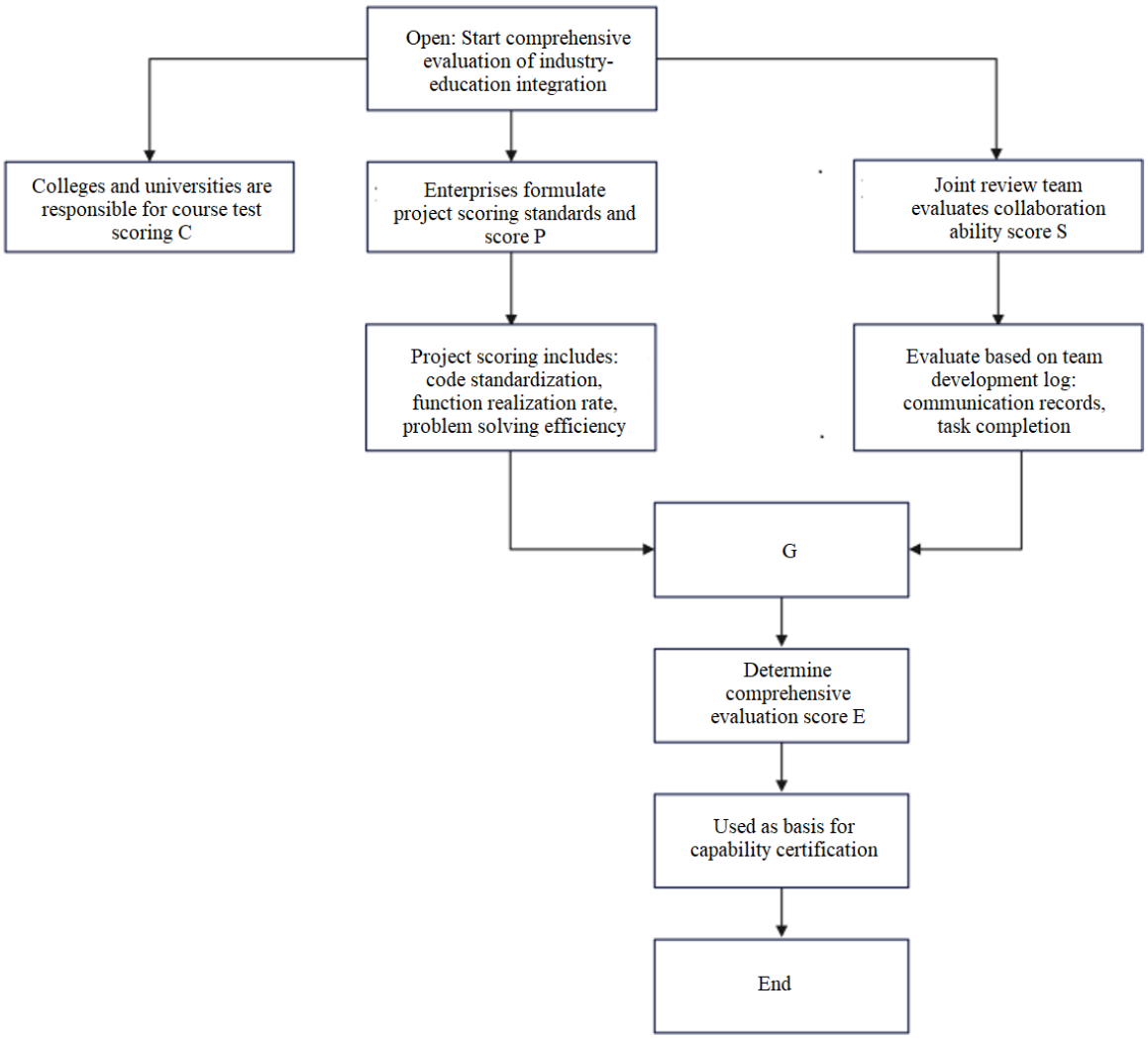


**Figure 4: Flow chart of comprehensive competency evaluation of industry-education integration**

## 4.5 Evaluation system for integration of industry and education

Design comprehensive ability evaluation formula: $E = \lambda_1 \times C + \lambda_2 \times P + \lambda_3 \times S$ (where E is the comprehensive evaluation score, C is the course test score, P is the enterprise project practice score, S is the collaborative ability evaluation score, $\lambda_1$, $\lambda_2$, $\lambda_3$ are weights and $\lambda_1 + \lambda_2 + \lambda_3 = 1$)). Enterprises participate in formulating the scoring criteria for P, including code standardization, function realization rate, and problem solving efficiency indicators; colleges and universities are responsible for the assessment of C, using a standardized test question bank; a joint review group is established to assess S, and scores are given based on the communication records and task assignment completion in the team development log. Substitute the scores of each item into the formula to calculate the E value, which is used as the basis for ability certification [6].

Table 6 shows the impact of course tests, enterprise project practice, and collaborative ability scores on the comprehensive evaluation score under different weights.

## 5 Conclusion

The foundation of cultivation is strengthened through the basic matrix, advanced module, Project practical training, dynamic mechanism and industry-education evaluation. It is necessary to adjust the dimension of competence, update the content and optimize the indexes with the change of technology and demand, so as to keep the system dynamically adapted to the positions and narrow the gap between supply and demand.

**Table 6: Comprehensive ability evaluation score statistics**

| Project | Weights ($\lambda_1$, $\lambda_2$, $\lambda_3$) | Exam Score (C) | Industry Project Score (P) | Collaboration Score (S) | Composite Score (E) |
|---|---|---|---|---|---|
| Project1 | 0.4, 0.3, 0.3 | 85 | 90 | 88 | 87.3 |
| Project2 | 0.3, 0.4, 0.3 | 90 | 85 | 92 | 88.6 |
| Project3 | 0.2, 0.5, 0.3 | 88 | 95 | 85 | 89.4 |
| Project4 | 0.5, 0.2, 0.3 | 92 | 80 | 90 | 87.4 |
| Project5 | 0.3, 0.3, 0.4 | 80 | 88 | 95 | 87.7 |

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Key Research Project for Educational Reform in Higher Education Institutions of Hainan Province in 2025: Integrated Curriculum Research and Practice on Web Full Stack Development Talent Cultivation under the Context of Industry-University Integration. Project Number: Hnjg2025ZD-82

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